INTRODUCTION TO ECONOMICS
MGT404E

Producing in Perfectly Competitive Industries

Yale

Recap and Today's Plan

Last Two Lectures

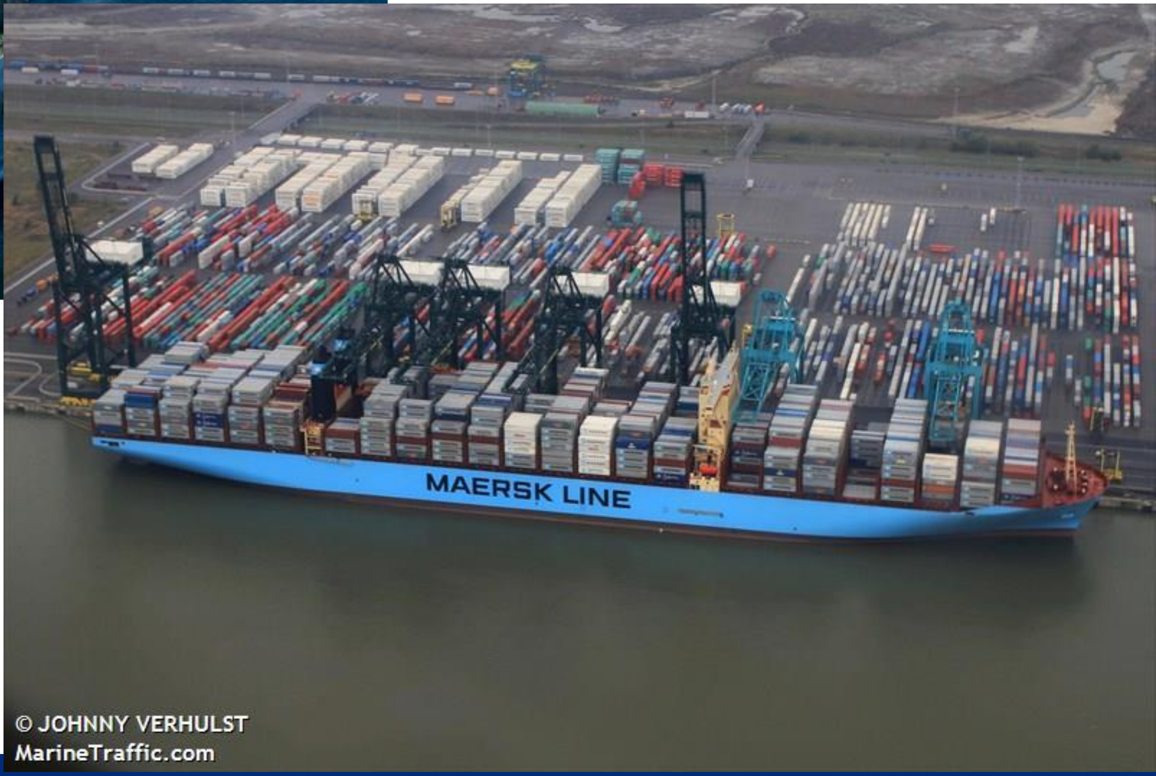
- The Production Function
- Cost Minimization
  - Finding "Marginal Product" and equating MPI to find optimal production function
  - Solving for inputs in terms of quantity, and then costs as a function of quantity
  - Taking the derivative to get "best" Marginal Cost
- Rebalancing production when things change
- Trade: theory of comparative advantage
- Trade policies tend to destroy surplus.

Today

Turning to firm decisions in….Perfectly Competitive Markets

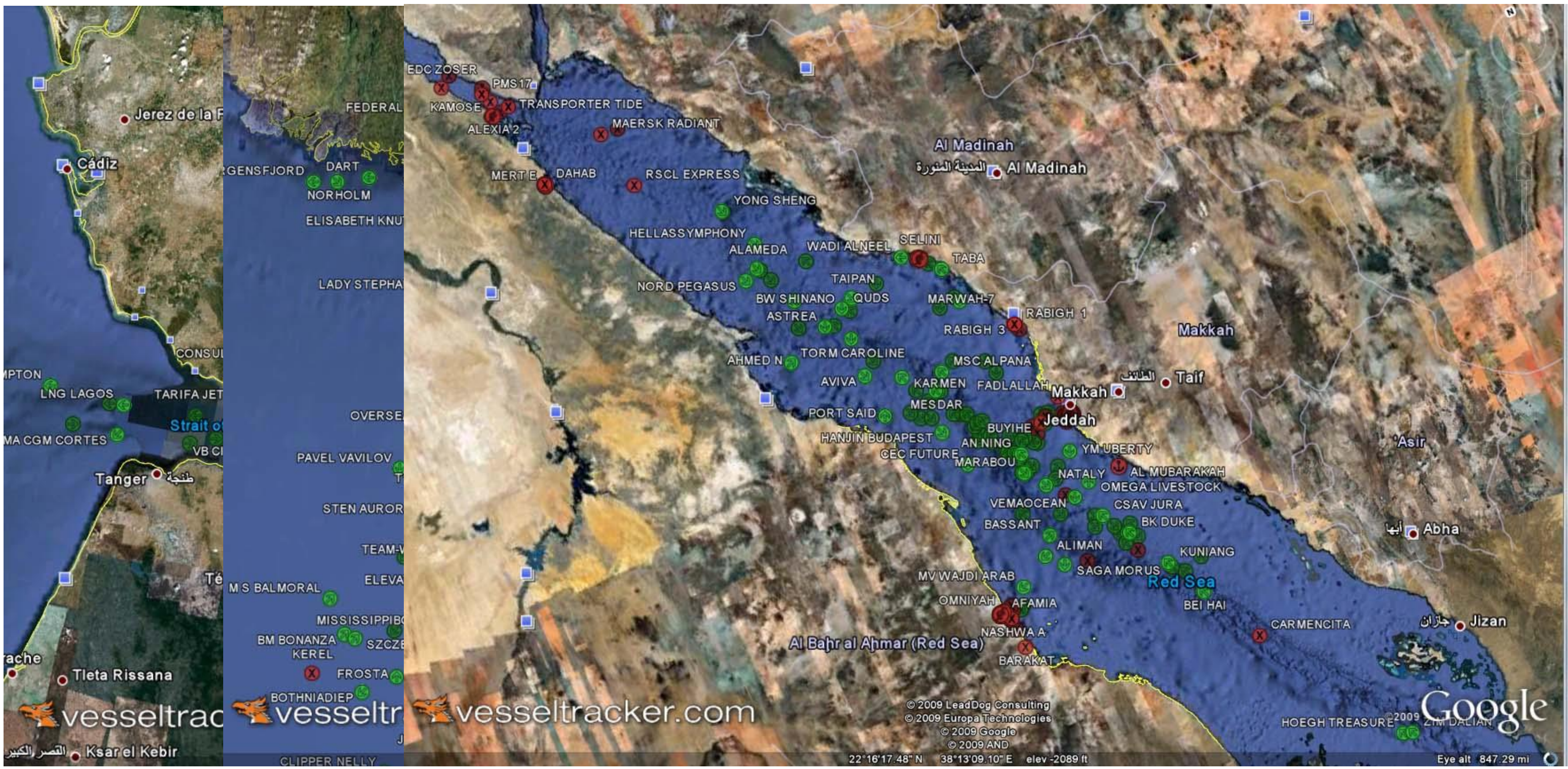
- Definition of Perfect Competition
- How do firms make decisions?
  - How much to produce?
  - Whether to produce?
- What does the market look like in the long run?

Shipping



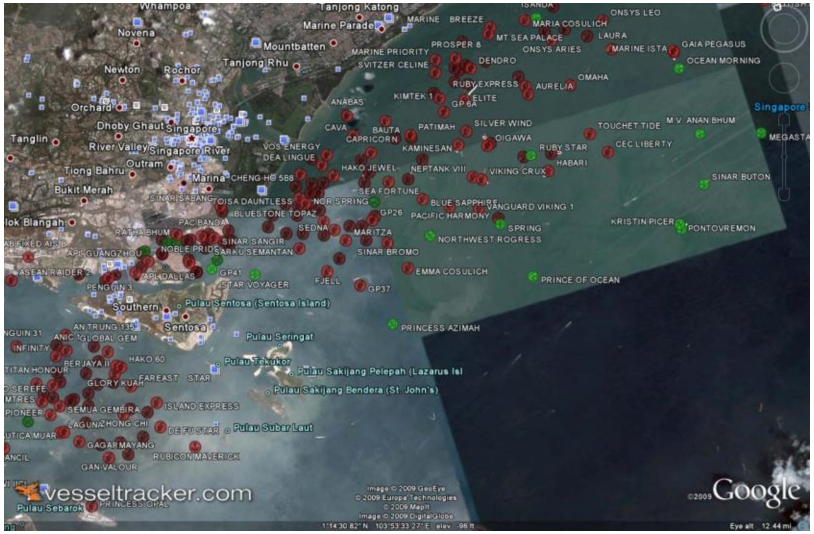
2009: The "Ghost Fleet" off Singapore

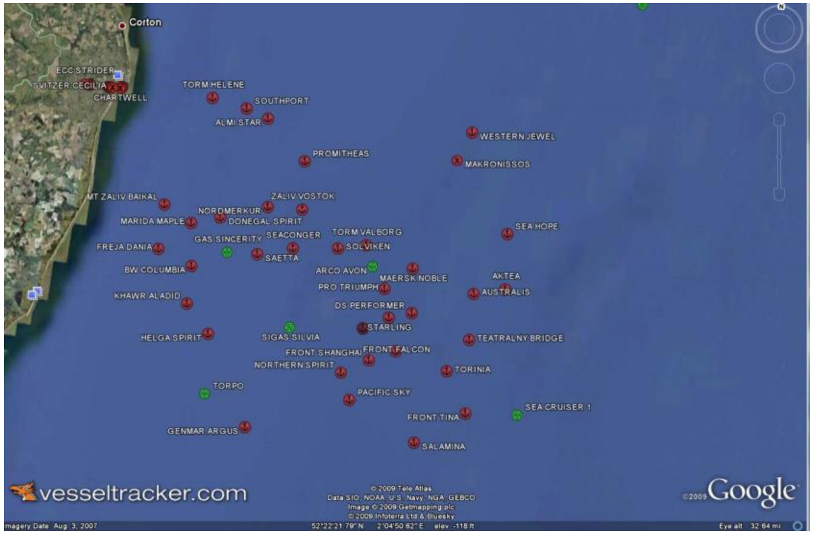
Data from vesseltracker.com (2009)



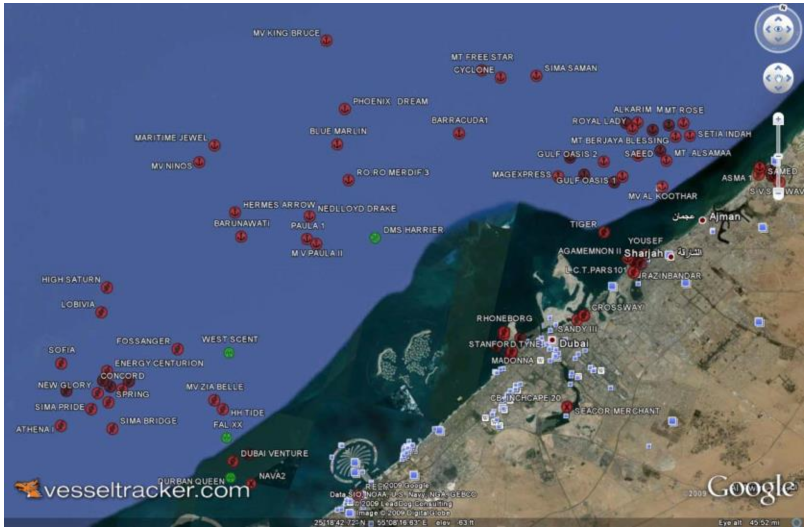
(Green indicates active ships, red indicates currently unused)
Many shipping channels very active during this time period…

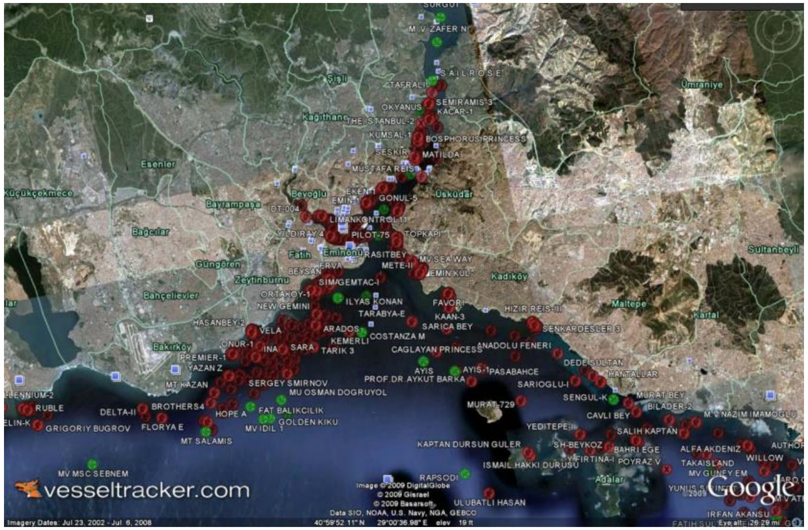
Data from vesseltracker.com (2009)

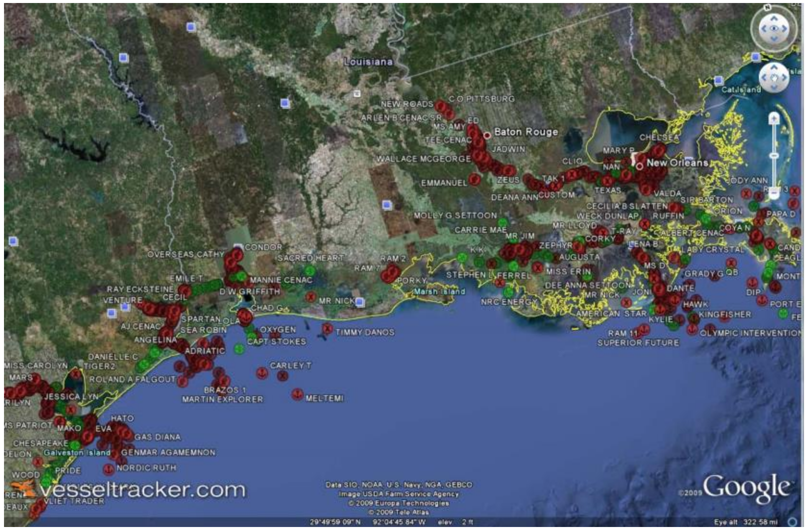




Data from vesseltracker.com (2009)







Today's Plan

Perfectly competitive markets

1. Perfect Competition

- **Conditions—reality and analogy**
- **What demand looks like**

2. Application

- Trends and startups
- Cupcakes, yogurt, home cleaning, car services, meal delivery…

3. Firm Behavior, Short Run

- Production decision: maximizing profits
- Set MR=MC, or P=MC
- Shutdown decision: if P<ATC, close the doors

4. Firm Behavior, Long Run; Market Equilibrium

- March to zero profits (entry and exit)
- Trying to escape perfect competition

What is Market Structure?

A market's structure provides information about how firms operating in the market will behave

Number/Size of Firms

Ease of Entry/Exit

Degree of Product Differentiation

Market Structure

- Perfect Competition?
- Oligopoly? Duopoly?
- Monopoly?

Today’s Class: Perfect Competition

What do a wheat farm, gas station, construction firm and lumber mill have in common?

- **Many small firms**
- **Easy to enter mkt**
- **Low differentiation**

Individual firms can’t influence the market. So, firms behave as if market characteristics are “given”.

Reality and analogy

- Has anyone worked in a market that had these characteristics?
- We’re also going to learn that firms in perfectly competitive markets want to escape these conditions.
- Perfect competition is a useful analogy for what happens when we are *close* to these conditions (many firms, e.g., restaurants), and a useful warning to firms of why differentiation and innovation matter

Perfect Competition

Assumption	Description	Implication
Undifferentiated products	• Consumers perceive each firm's product to be the same	**Law of One Price**: there is a single price at which transactions will occur
Perfect information	• Consumers know the prices being charged by all sellers	
Buyers are small	• A buyer cannot affect the market price	**Price Takers**: buyers and sellers take prices as given when making purchase and production decisions
Sellers are small	• A seller cannot affect the market price, nor input prices	
Identical firms (long run)	• Firms (and potential entrants) have access to the same technology (and hence production cost)	**Free Entry**: new firms will enter if it would be profitable

Perfect Competition

How realistic are these assumptions in typical markets?

- **Undifferentiated products?**
- **Perfect information?**
- **Buyers are small?**
- **Sellers are small?**
- **Identical firms?**

Perfect competition is a "limit case" that is useful to study, even if assumptions are strict

Possible Examples:
- Agriculture
- Commodities
- Construction

Today's Plan

Perfectly competitive markets

1. Perfect Competition

- Conditions—reality and analogy
- What demand looks like

2. Application

- **Trends and startups**
- **Cupcakes, yogurt, home cleaning, car services, meal delivery…**

3. Firm Behavior, Short Run

- Production decision: maximizing profits
- Set MR=MC, or P=MC
- Shutdown decision: if P<ATC, close the doors

4. Firm Behavior, Long Run; Market Equilibrium

- March to zero profits (entry and exit)
- Trying to escape perfect competition

Examples

Can you think of other markets that fit these assumptions?

- **Undifferentiated products?**
- **Perfect information?**
- **Buyers are small?**
- **Sellers are small?**
- **Identical firms?**

Cupcakes


BLEECKER ST
ONE WAY
The Magnolia Bakery
401
The Magnolia Bakery
batter
PCAKES

Evaluate a Startup Based on the Perfect Competition 5:

Assumption	Cupcakes
Undifferentiated products	• Easy to copy
Perfect information	• Easy to know
Buyers are small	• Yes!
Sellers are small	• None will get big with free entry
Identical firms	• There is no secret to making cupcakes…

Froyo, Cookie Dough, what else?

Today's Plan

Perfectly competitive markets

1. Perfect Competition

- Conditions—reality and analogy
- What demand looks like

2. Application

- Trends and startups
- Cupcakes, yogurt, home cleaning, car services, meal delivery…

3. Firm Behavior, Short Run

- **Production decision: maximizing profits**
- **Set MR=MC, or P=MC**
- **Shutdown decision: if P<ATC, close the doors**

4. Firm Behavior, Long Run; Market Equilibrium

- March to zero profits (entry and exit)
- Trying to escape perfect competition

Firm Decisions in The Short & Long Run

Maximizing profits under Perfect Competition:

Recall:

Firms cannot do anything about the price—that is set by the market.

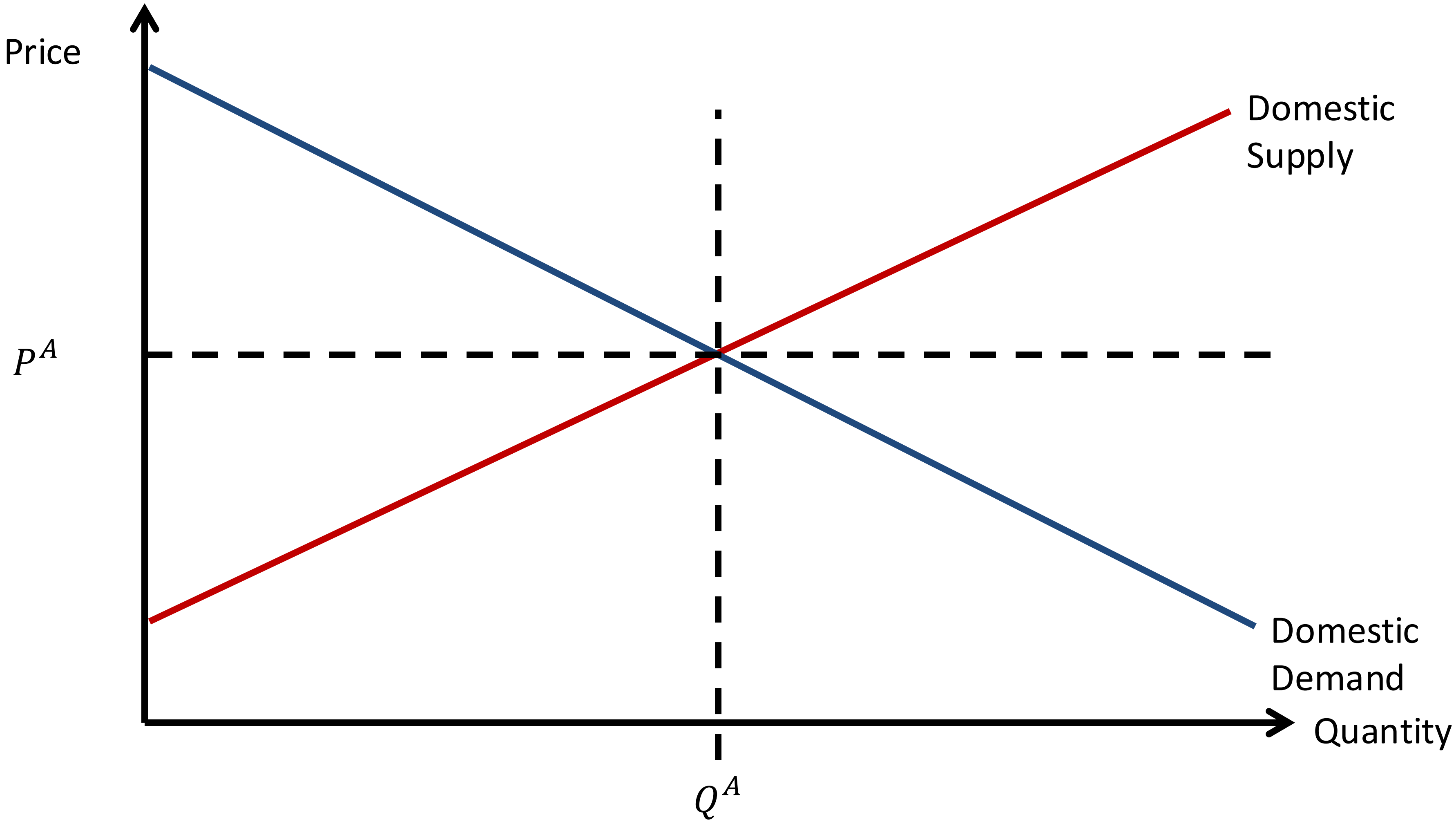
Firms can't cut costs in the short run and firms are identical in the long run.

Firms decide:

1. **Output decision**: If the firm produces, what output level (q^*) maximizes its profit or minimizes its loss?
   - This is a **short run** decision. Choose q, conditional on being in the market and having paid fixed costs.

2. **Exit decision**: Is it more profitable to produce q^* or to shut down and exit the market?
   - This is a **long run** decision. Choose (q=0) or (q* if q>0), allows the firm to "escape" fixed costs in the long run if business is unprofitable.

Market Demand and Supply



Perfect Competition: Residual Demand

What kind of demand curve do firms in Perfectly Competitive markets face?

If I raise my price…

- Due to perfect information and undifferentiated goods, consumers will purchase from other firms

If I lower my price…

- Due to perfect information and undifferentiated goods, consumers will purchase only from me

Demand curve is very near "flat" (perfectly elastic)

Firms are "**price takers**"

Note: Demand curve facing the *firm* is flat as firms are small; we call this *residual demand* after all other firms have supplied.

The aggregate market demand is downward sloping.

Short run decision: Choose output to maximize profits

How should a firm decide which output level to produce at?

Three (equivalent) output rules lead to the same output decision:

Choose output to maximize profit function	**Choose output to set marginal profit to zero**	**Choose output so that marginal revenue equals marginal cost**

You should produce a unit if you're being paid more than the cost to marginally make that unit.

Decision Rules #1 and #2: Maximizing Profits

"Horizontal" demand: Take price *P* as given

Choose quantity *q* to maximize profits:

Firm Output Decision

Revenue function:

$$R(q) = p \cdot q$$

Cost function:

$$C(q)$$

Profits:

$$\pi(q) = R(q) - C(q)$$

To maximize profits, find the q* where you can't earn more by making an extra unit → one more unit won't change your profit → derivative of profit=0:

$$\frac{d\pi(q^*)}{dq} = 0$$

Example – Profit Maximization

- Your cargo ship chooses q, how many container-miles to ship every month (in millions).
- You have fixed costs of $10K per month (salary, maintenance)
- Variable costs (in thousands) per million container-miles are $40q + q^2$ (going faster and weighing more burns more fuel)
- The prevailing market price per million container-miles is $50K

FC = 10

VC(q) = 40q+q²

P = 50

Perfect Competition Profit Maximization

Profit function (in thousands):

$$\pi(q) = (50q) - (10 + 40q + q^2)$$

Maximize by setting derivative equal to zero:

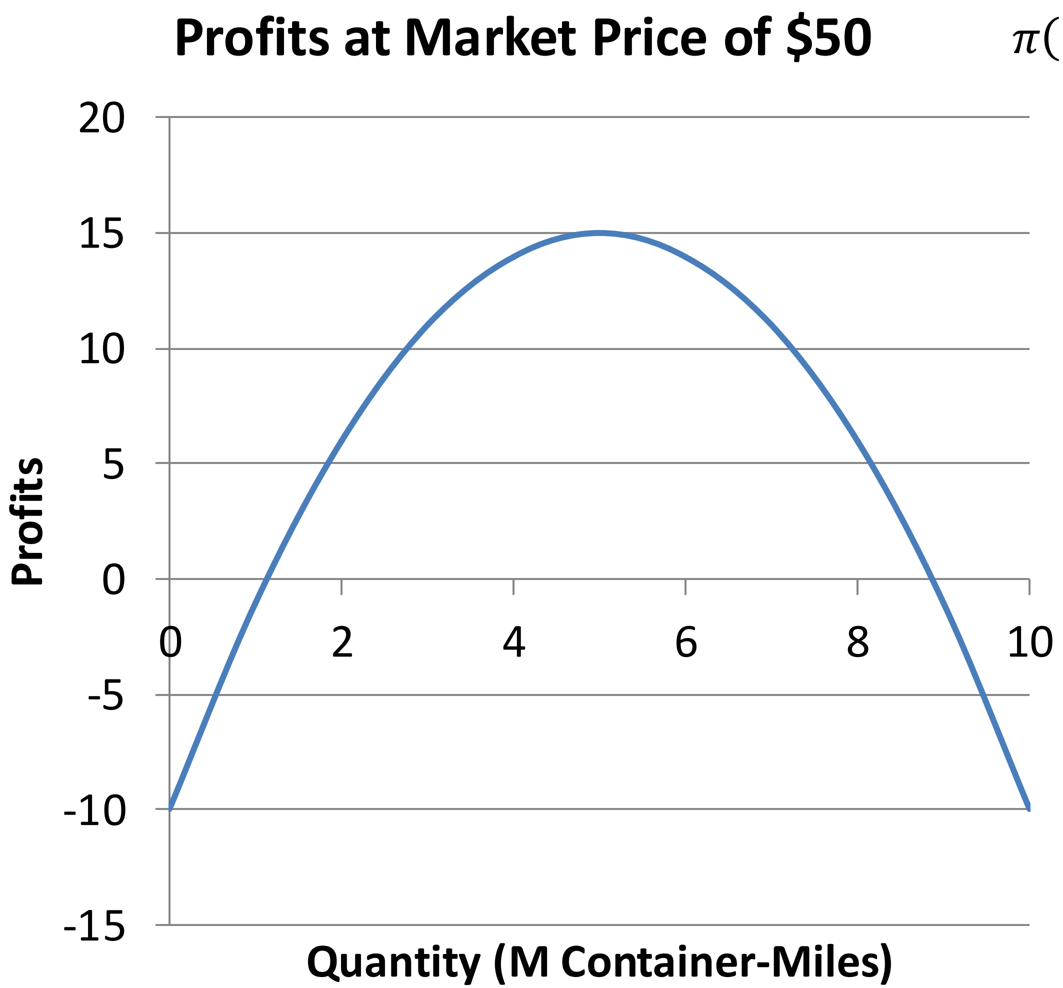
$$\pi'(q) = 50 - 40 - 2q$$
$$50 - 40 - 2q^* = 0$$
$$q^* = 5$$

So, the ship maximizes profits by sailing 5M container-miles this month, earning profits of $15,000.

$$\pi(5) = (50 \cdot 5) - (10 + 40 \cdot 5 + 25) = \$15$$

Example – Profit Maximization

$$\pi(q) = (50q) - (10 + 40q + q^2)$$



Sailing 5M container-miles maximizes profits for this ship.

Decision Rule #3: MR=MC

Firm Output Decision

Profits:

$$\pi(q) = R(q) - C(q)$$

Rules #1 and #2:

$$\frac{d\pi(q^*)}{dq} = 0$$

Marginal profits:

$$\frac{d\pi(q^*)}{dq} = \pi'(q) = R'(q) - C'(q) = MR(q) - MC(q)$$

Setting them equal to zero implies Rule #3: $MR(q) = MC(q)$

Decision Rule #3: MR=MC

Perfect Competition Profit Maximization

Profit equals revenue minus cost:

$$\pi(q) = R(q) - C(q)$$

So note that:

$$\pi'(q) = R'(q) - C'(q)$$

Meaning we maximize profit by setting:

$$R'(q^*) = C'(q^*)$$

Or,

$$MR(q^*) = MC(q^*)$$

Marginal Revenue = Marginal Cost

In our case, $R(q) = p \cdot q$

$$MR = p$$

$$MC(q) = \frac{dVC(q)}{dq} = 40 + 2q$$

We get that the profit maximizing quantity, in general, is given by

$$q^* = \frac{1}{2}p - 20$$

The cargo ship's optimal amount of container-miles to sail depends on the prevailing market price.

Equivalently, profits are maximized at q^* that sets p $=$ MC in perfect comp.
(Recall: MC curve = a firm's supply curve!)

Perf. Competitive Firm Behavior Depends on Price

Recall: Average Total Cost and Average Variable Cost

The **Average Total Cost** (ATC) is the total cost per unit of output. The **Average Variable Cost** (AVC) is the total variable cost per unit of output. The main distinction is that ATC includes fixed costs.

$$ATC = {}^{TC}/_{q} = {}^{F}/_{q} + {}^{VC}/_{q} \qquad\qquad AVC = {}^{VC}/_{q}$$

In the short run, fixed costs are unavoidable; firm will decide looking at the min(AVC).

Market Price	Outcome
$\boldsymbol{P > \min(ATC)}$	• Produce and earn a profit • Market price is above average cost
$\min(\boldsymbol{AVC}) < \boldsymbol{P} < \min(\boldsymbol{ATC})$	• Produce and earn a loss • Produce to cover fixed costs
$\boldsymbol{P < \min(AVC)}$	• Shut down production • Selling any quantity would increase losses

More about Shutting Down

FC = 10

$VC(q) = 40q+q^2$

P = 50

Shut Down Decision

Average Variable Cost (AVC):

$$AVC(q) = \frac{VC(q)}{q} = 40 + q$$

minimized when q $= 0, AVC(0) = 40.$

Average Total Cost (AVC):

$$ATC(q) = \frac{FC + VC(q)}{q} = \frac{10}{q} + 40 + q$$

minimized when-$\frac{10}{q2} + 1 = 0, q = \sqrt{10}, ATC\sqrt{10} = 46.32$

So, as long as the market price is above $40K per million container-miles, the ship will continue to operate.

Example – Price Ranges

Market Price	Outcome
$P > \$46.32$	• Cargo ship offers a positive quantity and earns positive profits. Follows $q^* = \frac{1}{2}p - 20$.
$P = \$46.32$	• Offers a positive quantity ($q^* = 3.16$) but earns zero profits: variable profits exactly cover fixed costs.
$\$40 < P < \$46.32$	• Offers a positive quantity but earns *negative* profits. • Fixed costs are unavoidable
$P \leq \$40$	• Shut down production, incurs fixed costs.

Today's Plan

Perfectly competitive markets

1. Perfect Competition

- Conditions—reality and analogy
- What demand looks like

2. Application

- Trends and startups
- Cupcakes, yogurt, home cleaning, car services, meal delivery…

3. Firm Behavior, Short Run

- Production decision: maximizing profits
- Set MR=MC, or P=MC
- Shutdown decision: if P<ATC, close the doors

4. Firm Behavior, Long Run; Market Equilibrium

- **March to zero profits (entry and exit)**
- **Trying to escape perfect competition**

Short Run vs Long Run

Recall:

When we discussed topics like elasticity, we found that firms/consumers are less flexible in the short run than in the long run.

Characteristic	Short Run	Long Run
Entry/Exit	• Number of firms in the industry is fixed	• Firms can enter or exit
Fixed Costs	• Fixed costs are unavoidable • i.e. sunk and cannot be recovered.	• Firms can avoid fixed costs by exiting

Long run *supply* is going to be more elastic than short run supply

Firm Behavior in the Long Run

Behavior	Description
Output Decision	• Same is in Short Run: choose q^* such that MR=MC (where MR=p, since firms are price-takers)
Entry/Exit Decision	• Firms can now enter/exit: firms enter if positive profits can be made; firms exit if $p < \min(ATC)$

Market Price	Short Run	Long Run
$\boldsymbol{P} > \min(\boldsymbol{ATC})$	• Produce and earn a profit; price > average cost	• Produce and earn a profit; price > average cost
$\min(\boldsymbol{AVC}) < \boldsymbol{P} < \min(\boldsymbol{ATC})$	• Produce and earn a loss to cover fixed costs	• Exit
$\boldsymbol{P} < \min(\boldsymbol{AVC})$	• Shut down production	• Exit

How is the market price determined in the long run?

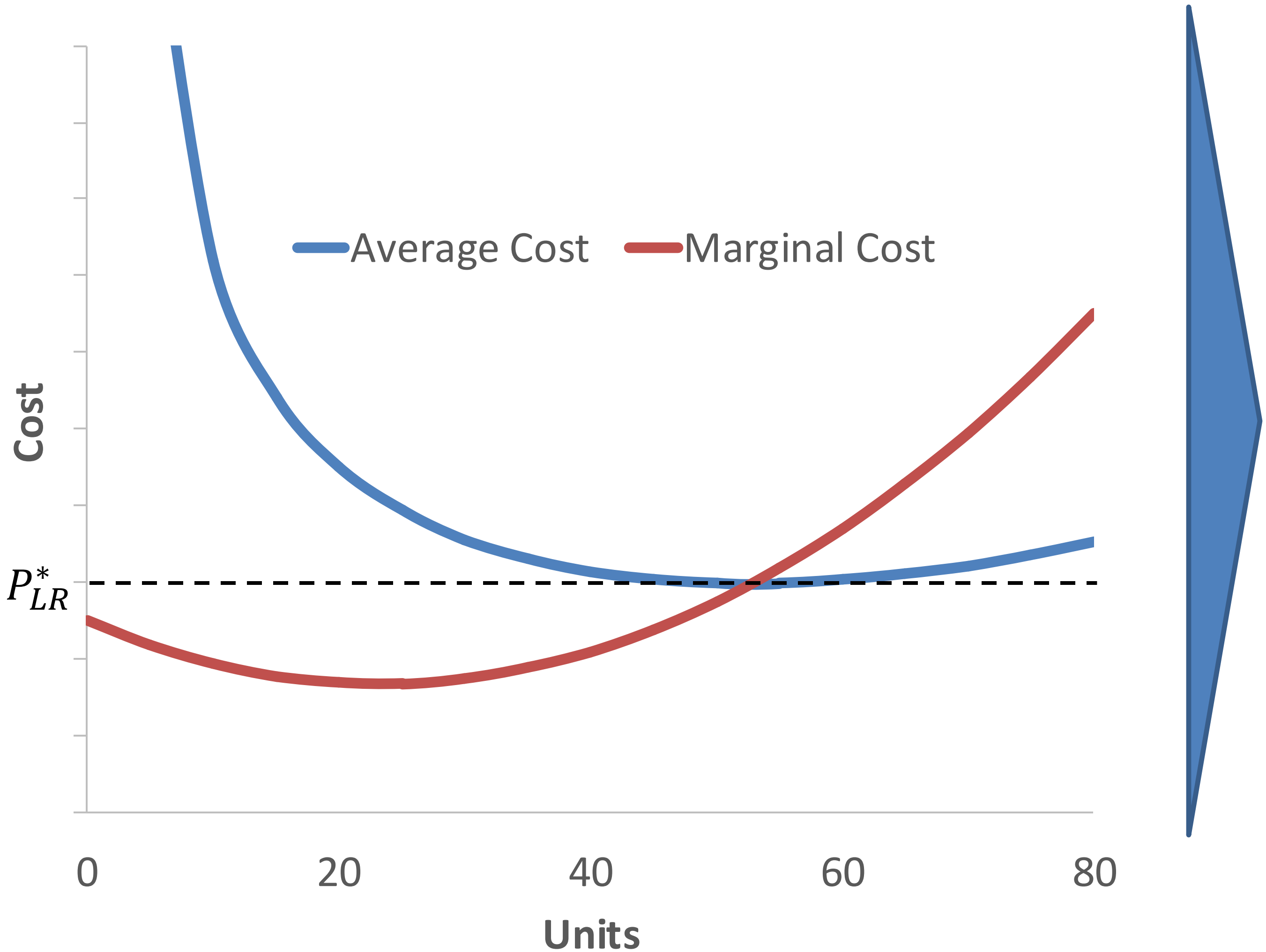
- If $p < \min(ATC)$, firms exit, driving price up
- If $p > \min(ATC)$, firms enter, driving price down
- When does this stop?
- What does this imply for profits?

Long run equilibrium in perfectly competitive markets:

p=min(ATC)

$$\boldsymbol{\pi(q) = 0}$$

Long Run Price in a Perfectly Competitive Market



- If we assume there is free entry and exit, the long run price is min(ATC)
- Above that: entry
- Below that: exit
- For our shipping example, this price is $46.32

Shipping Glut: What happened?

Demand:
Shipping service is undifferentiated, each ship acts as a price-taker, large fixed costs in short run.

Supply:
Huge number of ships ordered and built before economic downturn, long lead time

Result

- Each ship can be thought of as a "firm", faces very elastic demand
- Supply of ships is almost perfectly inelastic due to cost of scrapping, lead time to build
- Short run: "Ghost Fleet" bigger than the US and British navies combined sits idle off Malaysian coast.
- Long run: cargo ships scrapped, converted into other types of ships
  - Converted into cruise ships (WSJ, 10/8/08), run onto beaches for scrap.

Side note: how did shipping become undifferentiated?

- The development of the standardized shipping container happened in the 1950s-1970s. This standardization was opposed by shipping companies… (Read "**The Box**" for more)

Shipping Glut: Escaping Perfect Competition

- Super tankers like the *Mary Maersk* escape the perfect competition trap by being different
- In this case, they have a lower cost structure (a different *technology*)
- In other words, they can profitably operate at far lower prevailing prices

One upside: created a market for ship-shipping ships

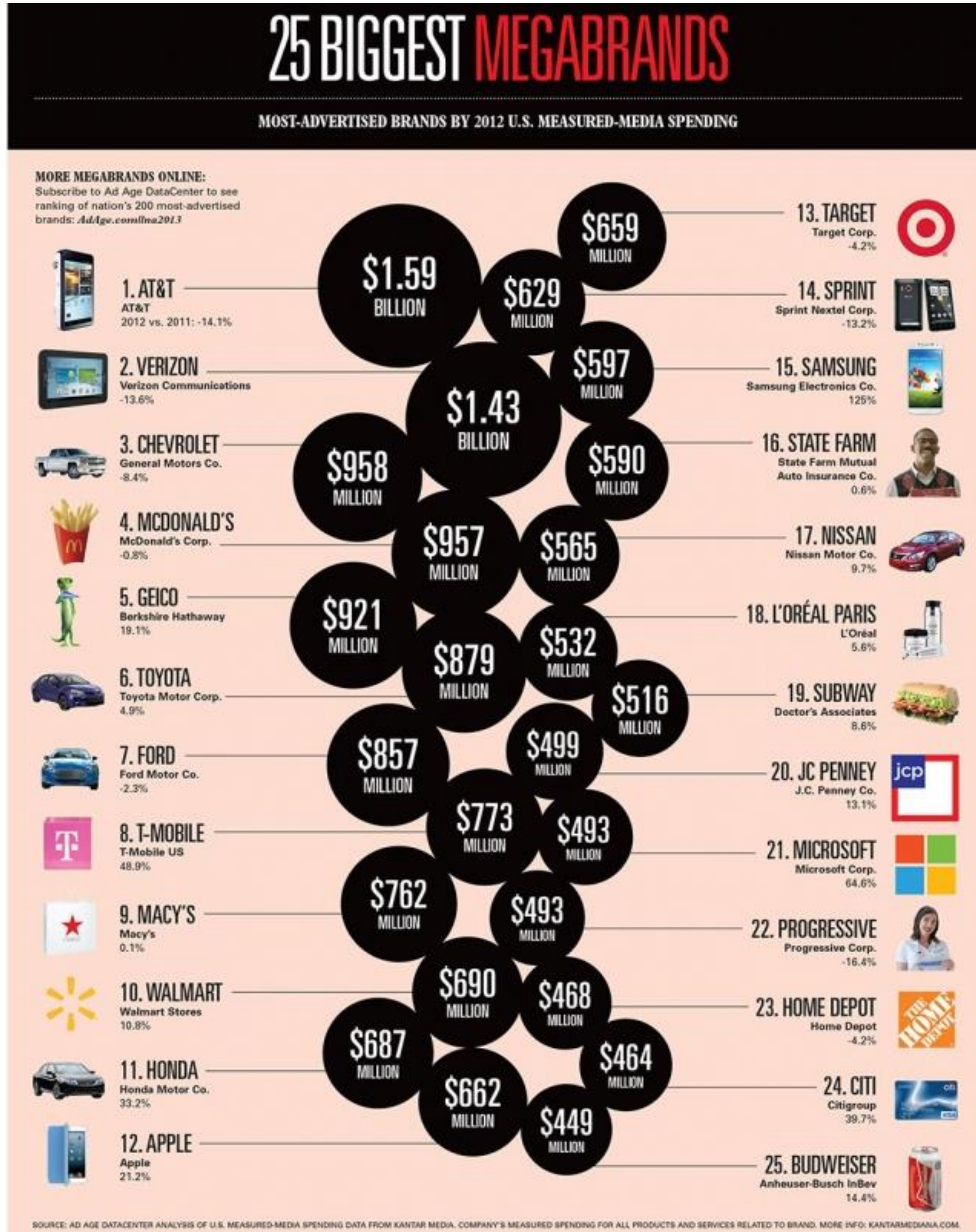
Can Firms Avoid Perfect Competition?

Getting Around the Assumptions

Assumption	Way firms try to dodge the assumptions
Undifferentiated products	Differentiate, marketing, improve quality.
Perfect information	Make pricing more opaque?
Buyers are small	(This one is good for the firm)
Sellers are small	Mergers, exclusive contracts, collusion/cartels (not recommended!!)
Identical firms, free entry	Develop a cost advantage. Achieve economies of scale or scope. Innovate and patent new technology. (often lobbying groups attempt to get favorable regulations)

Create a brand:

Convince consumers that competitors aren’t close substitutes after all



You can even differentiate your product overnight!

A product can be undifferentiated in one market but quite differentiated in another

What are these queues for?

Krispy Kreme!


The New York Times

Doughnut chains are rolling into China, Taiwan, South Korea and Japan

June 12, 2007

Summary & Direction

Today

- Perfectly competitive markets: firms have no market power, act as price-takers
- Profit maximization sets p=MC.
- Firms will produce at a loss in the short run if it helps them cover their fixed costs.
- In the long run, firms enter if there are profits to be made, and exit if they are incurring losses.
- Long run equilibrium has p=min(ATC) and no profits.

Next Lecture

- Firms with market power
- Setting a price with market power: tradeoffs from uniform pricing
- The Inverse Elasticity Pricing Rule